

Assignment #7 CSC 115

39.) Give the three representations of an XOR gate and say in words what XOR means.

Boolean Expression

$$X = A \oplus B$$

Logic Diagram Symbol



A XOR gate produces a 0 if 2 inputs are the same, + a 1 otherwise.

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

43.) Give the Boolean expression for a three input AND gate, and then show its behavior with a truth table.

$$X = A \cdot B \cdot C$$

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

46.) How does a transistor behave?

The transistor acts as a switch, are much smaller than vacuum tubes, + require less energy to operate. Also, able to switch states in nanoseconds.

58.) Draw a circuit diagram corresponding to the following Boolean expression:

$$A'B + (B+C)'$$

